

Task 1: Breast Cancer Ultrasound texture feature and context model

Goal

Build a model that extracts non-lesion ultrasound context features from the image:

- skin/superficial layer texture
- tissue composition texture
- lesion texture
- perilesional texture
- posterior shadow/enhancement region
- depth-dependent ultrasound information

This is not primarily segmentation. This is feature extraction.

Input

For each case:

- image.png
- tumor_mask.png
- classification label: benign/malignant/normal
- **optional metadata: pixel size, BI-RADS, echogenicity, posterior features, etc.**

Preprocessing

Each image should be converted to grayscale.

image: 256×256 or 384×384

mask: same size, nearest-neighbor interpolation

Normalize image intensity: $\text{image} = (\text{image} - \text{mean}) / \text{std}$

Create derived masks:

- tumor_mask = original lesion mask
- peritumoral_ring_5px = $\text{dilate}(\text{tumor_mask}, 5 \text{ pixels}) - \text{tumor_mask}$
- peritumoral_ring_15px = $\text{dilate}(\text{tumor_mask}, 15 \text{ pixels}) - \text{dilate}(\text{tumor_mask}, 5 \text{ pixels})$
- posterior_region = region below the tumor bounding box until bottom of image
- skin_region = top 10–15% of image height

- background_tissue_region = image area excluding tumor + peritumoral rings

Features to extract

A. Handcrafted texture features

For each region:

- tumor
- 5px peritumoral ring
- 15px peritumoral ring
- posterior region
- skin/superficial region
- background tissue

Extract:

- mean intensity
- std intensity
- entropy
- skewness
- kurtosis
- GLCM contrast
- GLCM homogeneity
- GLCM energy
- GLCM correlation
- LBP histogram features
- edge density
- shadow index = posterior mean intensity / tumor mean intensity
- lesion-to-background contrast

Output one CSV: case_id, region, feature_1, feature_2, ..., classification

B. Deep texture embedding

Use a pretrained encoder: ResNet50 / ConvNeXt-Tiny / DINOv2-small

Input should be cropped patches:

- tumor crop
- tumor + peritumoral ring crop
- posterior crop

- whole image

For each crop, extract embedding:

- tumor_embedding: 512 or 768 dimensions
- perilesion_embedding
- posterior_embedding
- whole_image_embedding

Save:

- features_deep.csv
- features_texture.csv
- features_combined.csv

Train simple classifiers first:

- Random Forest
- XGBoost / LightGBM
- Logistic Regression
- SVM
- MLP

Prediction target: benign vs malignant: Later optional targets:

- BI-RADS
- posterior_features
- echogenicity
- shape
- margin

Required output

- 1_preprocess.py
- 2_extract_texture_features.py
- 3_extract_deep_embeddings.py
- 4_train_classifier.py
- features_texture.csv
- features_deep.csv
- features_combined.csv
- results.csv

Metrics:

- AUC
- accuracy
- sensitivity
- specificity
- F1-score
- confusion matrix

Task 2 : Breast Cancer Ultrasound Pseudo-Context Segmentation and Region Discovery Model

Goal

Build a model that automatically generates ultrasound context maps from breast ultrasound images.

This task is different from Task 1. Task 1 will extract handcrafted and deep features from known lesion-derived regions. Task 2 will create pseudo-segmentation maps that identify different ultrasound context regions from the same image.

The aim is to approximate what a radiologist visually understands from ultrasound:

- skin/superficial layer
- breast tissue background
- tumor region
- peritumoral region
- posterior acoustic region
- possible posterior shadowing
- possible posterior enhancement
- deep tissue region

This is not ground-truth anatomical segmentation. This is pseudo-context segmentation and region discovery.

Input

For each case:

- image.png
- tumor_mask.png
- classification label: benign/malignant
- optional metadata: pixel size, BI-RADS, echogenicity, posterior features, etc.

Required Output Per Case

For every image, generate: - context_map.png - overlay_context_map.png - region_features.csv

The context map should use these labels:

- 0 = background / unknown

- 1 = skin or superficial layer
- 2 = background breast tissue
- 3 = tumor
- 4 = peritumoral region
- 5 = posterior acoustic region
- 6 = likely posterior shadow
- 7 = likely posterior enhancement
- 8 = deep tissue region

Step 1: Preprocessing

Each image should be converted to grayscale.

Resize:

- image: 256×256 or 384×384
- mask: same size using nearest-neighbor interpolation

Normalize image intensity:

$\text{image} = (\text{image} - \text{mean}) / \text{std}$

Also save a min-max normalized version for visualization:

$\text{image_visual} = (\text{image} - \text{min}) / (\text{max} - \text{min})$

Step 2: Tumor Anchor Extraction

Use the tumor mask to calculate:

- tumor bounding box
- tumor centroid
- tumor width
- tumor height
- tumor area
- upper boundary of tumor
- lower boundary of tumor
- left and right limits of tumor

These values will define all other context regions.

Step 3: Rule-Based Context Map Generation

Create basic pseudo-regions using the tumor mask.

###Tumor region

Use the original tumor mask:

tumor_region = tumor_mask

Peritumoral region

Create a ring around the tumor:

peritumoral_region = dilate(tumor_mask, 15 pixels) - tumor_mask

Skin or superficial region

Define the upper 10–15% of the image height:

skin_region = top 10–15% of image

Posterior acoustic region

Define the vertical region below the tumor bounding box:

posterior_region = area below tumor lower boundary until bottom of image, limited horizontally around tumor width with some margin

Deep tissue region

Define the lower one-third of the image:

deep_region = bottom 33% of image

Background breast tissue

Everything that is not tumor, not peritumoral ring, not posterior region, and not skin region:

background_tissue_region = image area excluding other regions

Step 4: Posterior Shadow and Enhancement Detection

Use intensity comparison below the lesion.

For each case, calculate:

- tumor_mean_intensity
- posterior_mean_intensity
- lateral_same_depth_mean_intensity

The lateral same-depth region means tissue at the same depth as the posterior region but left and right of the tumor.

Calculate: $\text{shadow_index} = \text{posterior_mean_intensity} / \text{lateral_same_depth_mean_intensity}$

Rules: If posterior_mean_intensity is clearly lower than lateral_same_depth_mean_intensity:

- label as likely posterior shadow

If posterior_mean_intensity is clearly higher than lateral_same_depth_mean_intensity: - label as likely posterior enhancement

Suggested thresholds:

- shadow if $\text{shadow_index} < 0.85$

- enhancement if shadow_index > 1.15
- otherwise no posterior acoustic change

These thresholds must be tested and adjusted after visual inspection.

Step 5: Superpixel-Based Tissue Region Discovery

Apply SLIC superpixel segmentation to the ultrasound image.

For each superpixel, extract:

- mean intensity
- standard deviation
- entropy
- LBP texture features
- GLCM contrast
- GLCM homogeneity
- depth coordinate
- distance from tumor centroid
- whether it overlaps tumor/peritumoral/posterior/skin region

Then cluster the superpixels using:

- k-means clustering
- Gaussian mixture model

Try cluster numbers: - 3 clusters - 4 clusters - 5 clusters

Save the best visual result.

Output:

- superpixel_cluster_map.png
- superpixel_cluster_overlay.png

Step 6: Deep Embedding-Based Region Discovery

Use one pretrained feature extractor:

- ResNet50
- ConvNeXt-Tiny
- DINOv2-small, if available

Divide the image into patches, for example:

- 16×16 patches
- 32×32 patches

Extract deep embeddings for each patch.

Cluster patch embeddings using:

- k-means clustering
- PCA or UMAP for visualization

Generate:

- deep_cluster_context_map.png
- deep_cluster_overlay.png

This output should show whether deep features can separate tissue-like regions without manual labels.

Step 7: Final Pseudo-Context Map

Combine:

- rule-based context map
- posterior shadow/enhancement map
- superpixel cluster map
- deep embedding cluster map

Create one final context map per image.

Priority order:

1. tumor
2. peritumoral region
3. posterior shadow/enhancement
4. skin region
5. posterior region
6. deep tissue region
7. background tissue

Save:

- final_context_map.png
- final_context_overlay.png

Step 8: Train a Context-UNet Student Model

After pseudo-context maps are generated for all images, train a U-Net-like model to predict the pseudo-context maps directly from the ultrasound image.

Input:

- image.png

Target:

- final_context_map.png

Architecture:

- Encoder: ResNet34, EfficientNet-B0, ConvNeXt-Tiny, or Swin-Tiny
- Decoder: U-Net decoder
- Output channels: 8 or 9 classes

Loss:

- CrossEntropy loss
- Dice loss

Training split:

- 70% training
- 15% validation
- 15% testing

Important: Do not call this true anatomical segmentation. It is pseudo-context segmentation.

Step 9: Evaluation Because there are no real ground-truth labels for these context regions, evaluation must include both quantitative and visual checks.

Quantitative metrics

For the model:

- pixel accuracy against pseudo-labels
- mean IoU against pseudo-labels
- Dice score against pseudo-labels

Visual quality control

Save overlay images for at least 100 random cases.

Each overlay should show:

- original ultrasound image
- tumor mask
- context map
- final overlay

Manual visual inspection should check:

- tumor region is correctly preserved
- peritumoral region surrounds tumor
- posterior region is below tumor

- skin region is at the top
- shadow/enhancement labels are visually plausible

Step 10: Final Deliverables

must submit:

- 1_generate_rule_based_context_maps.py
- 2_detect_shadow_enhancement.py
- 3_superpixel_region_discovery.py
- 4_deep_embedding_region_discovery.py
- 5_create_final_context_maps.py
- 6_train_context_unet.py
- 7_evaluate_context_unet.py
- final_context_maps/
- overlay_context_maps/
- superpixel_cluster_maps/
- deep_cluster_maps/
- context_region_features.csv
- context_unet_results.csv
- README.md explaining how to run the full pipeline

Expected Final CSV

The file context_region_features.csv should include:

- case_id
- image_filename
- classification
- tumor_area
- tumor_centroid_x
- tumor_centroid_y
- peritumoral_area
- posterior_mean_intensity
- lateral_same_depth_mean_intensity
- shadow_index
- posterior_label: none/shadow/enhancement
- skin_mean_intensity
- background_tissue_mean_intensity

- deep_tissue_mean_intensity
- number_of_superpixel_clusters
- dominant_background_cluster
- dominant_posterior_cluster

Final Purpose

This task will create an ultrasound context representation for each image.

Later, these outputs can be combined with Task 1 outputs:

- handcrafted texture features
- deep texture embeddings
- pseudo-context maps
- context-region features

Together, they will support a stronger downstream model for:

- benign/malignant classification
- BI-RADS prediction
- posterior feature prediction
- echogenicity prediction
- axillary lymph node metastasis prediction, if ALNM labels are available